The General Epistemic Dispersion Theory

Structural Sufficiency and the Separation of Global Validation from Configuration Reconstruction

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Abstract

This work develops the General Epistemic Dispersion Theory (GEDT) as a framework for analyzing how persistent structural information constrains global configurations in fragmented systems. GEDT distinguishes complete configurational identification from determination of a global verification outcome.

A reusable structural core S may leave positive residual uncertainty about the realized configuration Π while nevertheless being sufficient to determine the global outcome Y, so that $H(Y \mid F, S) = 0$ can coexist with $H(\Pi \mid F, S) > 0$.

This separation defines identifying and non-identifying regimes and motivates structural epistemic dispersion as the allocation of configurational uncertainty between persistent structure and dynamic information.

Using established information-theoretic tools, the framework further relates residual configurational uncertainty to the additional transcript required for reconstruction.

Reductions in communication, coordination, or amortized cost are treated as possible consequences of structural sufficiency rather than as the theory's primary contribution.

Keywords: structural epistemic dispersion; information theory; distributed systems; global integrity; side information; structural information; conditional entropy.

Introduction

A general problem in the design of distributed systems concerns the information cost required to determine global integrity from a set of partial data.
A broad class of distributed architectures addresses this limitation by sharing among nodes the information required for semantic and relational understanding of the entire system state from the partial data. This approach entails a significant information cost, as it may require the repeated transmission, replication, storage, or reconstruction of explicit metadata to resolve local uncertainty.

This manuscript introduces General Epistemic Dispersion Theory, an analytical framework aimed at quantifying the distribution of the information cost required for the reconstruction

and global validation of fragmented data. The framework formalizes the distinction between architectures based on explicit replication of semantic understanding and architectures in which some of the relations required for global understanding are embedded in a reusable structural core.

By applying established principles of information theory—specifically, conditional entropy, source coding bounds, and the Fano inequality—the paper demonstrates how the persistent structural core of a system can function as implicit metadata. In the presence of structural sufficiency, some or all of the information required for configurational reconstruction or global integrity determination can be transferred from reiterated dynamic communication to a persistent and reusable structure.
This reallocation reduces the residual uncertainty that must be resolved with an additional transcript.

It can yield benefits in terms of communication, coordination, latency, and energy when the cost of building, distributing, accessing, and maintaining the structure is lower than the cost of the dynamic operations it avoids.

Original contribution and theoretical scope

The General Epistemic Dispersion Theory (GEDT) does not propose new elementary principles of information theory, nor does it claim conditional entropy, mutual information, source-coding bounds, or the Fano inequality as original mathematical results. These established tools provide the formal basis for addressing a different theoretical question: **how much of the global configuration of a fragmented system must be epistemically resolved in order to determine a global property of that system?**

GEDT distinguishes two tasks that are often treated together but need not require the same information: **complete configurational identification** and **global validation**. Let Π denote the realized global configuration, F the observed fragments, S a persistent structural core, and Y a deterministic global verification outcome.
Complete identification requires the uncertainty about Π to vanish,

$$H(\Pi \mid F, S) = 0,$$

whereas exact global validation requires only

$$H(Y \mid F, S) = 0.$$

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Because Y is a function of the global configuration together with the available fragments and structural information, these two conditions are not equivalent.
GEDT explicitly admits the regime:

$H(Y \mid F, S) = 0$ while $H(\Pi \mid F, S) > 0$,

in which the global verification outcome is exactly determined although the underlying global configuration remains non-identifiable.

The theory therefore separates **epistemic sufficiency for a global task** from **epistemic completeness with respect to the global state**.

Within this framework, S is not introduced merely as generic side information. It represents a persistent and reusable structural core containing feasibility constraints, invariants, structural relations, role schemas, and decoding geometry that restrict the set of configurations compatible with the observed fragments. Its informational contribution is quantified by

$D_{epi}(S ; \Pi \mid F) = I(\Pi ; S \mid F)$,

while the residual configurational uncertainty is

$H_{res} = H(\Pi \mid F, S)$.

Hence,

$H(\Pi \mid F) = D_{epi}(S ; \Pi \mid F) + H_{res}$.

The underlying identity is standard.
The theoretical proposal lies instead in using this decomposition to characterize how persistent structural constraints redistribute the epistemic burden associated with global configuration, and in distinguishing the information required to identify the configuration from the information required to determine a specified global outcome.

GEDT consequently distinguishes **identifying regimes**, in which $H_{res} = 0$, from **non-identifying regimes**, in which multiple global configurations remain compatible with F and S. Importantly, a non-identifying regime need not imply uncertainty about global validity: structurally different configurations may remain epistemically unresolved while being equivalent with respect to the verification outcome.

The framework further introduces an additional configurational transcript M_{Π} and relates its expected informational length to residual configurational uncertainty:

$E[\ell(M_\Pi)] \geq \max\{0, H_{res} - \delta_{\mathcal{P}}(\epsilon_\Pi)\}$.

This bound quantifies the additional information required when the persistent structural core does not itself make the configuration sufficiently identifiable at the prescribed error level.

The primary contribution of GEDT is therefore **not the general observation that side information can reduce uncertainty, nor that persistent structure may reduce repeated communication**. Rather, it is the proposed theoretical architecture for distinguishing and quantifying three different objects: the configurational information absorbed by reusable structure, the configurational uncertainty that remains unresolved, and the potentially smaller amount of information required to determine a global verification outcome.

Communication, coordination, latency, energy, or amortized economic advantages may follow in particular implementations when persistent structure replaces information that would otherwise have to be supplied dynamically. Such advantages are treated as **secondary system-level consequences** of structural epistemic sufficiency and are not assumed to constitute the fundamental theoretical novelty of GEDT.

Positioning Relative to the Strongest Prior Paradigms

GEDT lies at the intersection of several established traditions in information theory, statistical inference, distributed computation, structural representation, and knowledge preprocessing. Its proposed contribution therefore does not depend on claiming novelty for any single mathematical ingredient used in the framework.

In particular, GEDT does not claim as original the facts that side information can reduce conditional uncertainty; that a function may be determined without reconstructing the complete underlying source; that a sufficient representation may preserve information relevant to a specified inferential or decision task while discarding other distinctions; that equivalence classes may provide minimal representations for prediction; that information may be separated between a model and a residual description; or that a reusable preprocessed representation may reduce repeated computational or communication costs. These principles have substantial precedents in the literatures discussed below.

The proposed contribution of GEDT instead concerns their organization within a single configurational framework in which a persistent reusable structure S, a realized global configuration Π, residual configurational uncertainty, task-specific global validation Y, and execution-specific transcripts are treated as distinct epistemic objects.

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Side information and correlated-source coding

Classical source-coding results show that information already available to a decoder can reduce the additional information required for reconstruction. In particular, the Slepian–Wolf framework establishes coding limits for correlated sources when correlated information is available across the distributed coding problem (Slepian & Wolf, 1973).
Related zero-error formulations also show that side information can induce combinatorial structure on the set of distinguishable source states (Witsenhausen, 1976).

GEDT does not claim this reduction of conditional uncertainty as novel. Formally, S may act as side information and reduce

$H(\Pi \mid F)$ to $H(\Pi \mid F, S)$.

Its proposed role is more specific, however.
S is modeled as a persistent and reusable structural core containing feasibility constraints, invariants, relations, role schemas, and decoding geometry that may be shared across multiple compatible configurations or executions. GEDT then separates the configurational information associated with such reusable structure from the uncertainty that remains instance-specific.

Functional computation and task-dependent equivalence

A particularly close precedent is the literature on functional computation. In *Coding for Computing*, Orlitsky and Roche consider communication when the receiver seeks to evaluate a function of distributed data rather than reconstruct the complete source, and characterize communication requirements through graph-based informational quantities (Orlitsky & Roche, 2001).
Earlier work by Han and Kobayashi (1987) also characterized classes of functions of correlated sources for which function computation and source reconstruction have fundamentally different informational requirements.
Accordingly, GEDT does not claim as novel the general proposition that a function can sometimes be determined without complete reconstruction of the underlying state. Nor is the inequality

$H(Y \mid F, S) \leq H(\Pi \mid F, S)$

presented as a new elementary information-theoretic law.
GEDT incorporates this established distinction into a different configurational architecture. For each realized pair (f, s), the conditional feasible support

$P_{f,s} = \mathrm{supp}(\Pi \mid F=f, S=s)$

represents the configurations that remain epistemically possible after conditioning on the fragments and persistent structure. The verification function induces an equivalence relation on this support,

$\pi \sim_{f,s} \pi' \Leftrightarrow \mathcal{V}(f, \pi, s) = \mathcal{V}(f, \pi', s)$.

GEDT therefore distinguishes the uncertainty required to identify which configuration actually occurred from the uncertainty required merely to distinguish configurations belonging to different global-validation classes. This distinction is subsequently connected to reusable structural information and execution-specific transcript requirements within the same model.

Finally, classical notions of common information provide a further precedent for distinguishing information genuinely shared across variables from total mutual dependence (Gács & Körner, 1973). GEDT does not identify its canonical structural core with Gács–Körner common information.

Sufficient statistics and decision-relevant information

The idea that complete raw information need not be retained for a specified inferential purpose is classical. Statistical sufficiency identifies representations that retain the information required for inference, while Blackwell's comparison of experiments formalizes the usefulness of information relative to decision problems (Fisher, 1922; Blackwell, 1953).

GEDT therefore does not claim the general concept of task-specific sufficiency as original. Its structural variable S, however, is not introduced merely as a statistic computed from the current observation. It is a persistent component of the system architecture that restricts the feasible configurational support across compatible instances.

The GEDT question is consequently not only whether an observation can be compressed without losing information about a parameter or decision, but how the epistemic burden of a fragmented global configuration is distributed between reusable structural constraints, residual instance-specific uncertainty, and the information needed for a specified global validation task.

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Information Bottleneck

The Information Bottleneck framework explicitly distinguishes information about an input from information relevant to a target variable and seeks compressed representations that retain task-relevant information (Tishby et al., 1999).

This is conceptually close to the validation–identification distinction in GEDT. Consequently, GEDT does not claim as novel the broad principle that some distinctions in an underlying state may be irrelevant to a specified target.

The objectives are nevertheless different. Information Bottleneck typically treats the representation itself as the object to be optimized under a compression–relevance trade-off. GEDT instead begins with a fragmented configuration and an explicitly modeled persistent structural core S, asks how that structure restricts the feasible configuration space, and separately quantifies residual uncertainty about Π and uncertainty about the global outcome Y.

The validation gap $\Gamma_{val} = H(\Pi \mid F, S) - H(Y \mid F, S)$,

then identifies the configurational uncertainty that need not be resolved merely to determine the specified global verification outcome.

Computational mechanics and predictive equivalence classes

Computational mechanics provides another relevant precedent. Causal states group histories that induce the same predictive distribution and yield minimal representations consistent with optimal prediction (Shalizi & Crutchfield, 2001).

GEDT therefore does not claim the general use of equivalence classes or minimal task-sufficient representations as an original construction. Its equivalence classes have a different object and purpose: they partition feasible global configurations, conditional on F and S, according to the value of an explicit global verification map.

Thus, computational mechanics concerns predictive equivalence of histories with respect to future distributions, whereas GEDT concerns validation equivalence of simultaneously feasible configurations with respect to a specified global property. The connection is conceptually important and limits any novelty claim based solely on equivalence-class reduction.

Algorithmic statistics and model–residual decompositions

Algorithmic statistics studies the relation between an individual data object and a model that captures its regularities, including two-part descriptions that distinguish structured model information from information remaining in the data (Gács et al., 2001).

This provides a clear precedent for the general intuition that information can be distributed between reusable structure and residual specification.
GEDT does not claim this intuition as new.

The GEDT formulation differs in using ensemble-based Shannon quantities and in explicitly separating persistent structural information from execution-specific configurational uncertainty in fragmented systems. It further distinguishes the residual information required for complete configurational identification from the potentially smaller amount required for a specified global validation task.

Knowledge compilation, preprocessing, and amortized information cost

Knowledge-compilation frameworks explicitly trade an up-front transformation or representation cost for more tractable repeated queries and transformations (Darwiche & Marquis, 2002). Likewise, information-complexity results relate information revealed by protocols to amortized communication complexity (Braverman & Rao, 2014).

GEDT therefore does not claim as novel the general principle that preprocessing or persistent representation can reduce repeated computational or communication burden. Its economic and amortized-cost results are secondary consequences of the framework.

The proposed theoretical role of S is instead to represent persistent structural constraints that reshape the conditional feasible configuration space itself. Possible savings in communication, computation, coordination, latency, or economic cost arise only when that epistemic restructuring has an advantageous implementation-specific cost.

Distributed snapshots, consensus, and global-state determination

Distributed snapshot algorithms address the acquisition of a consistent global state from distributed local information, whereas Byzantine agreement and consensus frameworks address coordinated decision making under faults or adversarial behavior (Chandy & Lamport, 1985; Lamport et al., 1982; Fischer et al., 1985).

GEDT addresses a different epistemic question. Local components need not themselves reconstruct the global configuration or reach agreement about it. They may produce only local observations or results, while global validation depends on those outputs together with a persistent structural core.

These paradigms are therefore relevant architectural neighbors, but they are less direct precedents for the validation–identification separation than functional computation, statistical sufficiency, Information Bottleneck, and task-equivalence approaches.

Scope of the proposed contribution

After these precedents are taken into account, the proposed contribution of GEDT is deliberately narrower than the novelty of its individual mathematical ingredients.

GEDT proposes a unified architecture in which: (F, S),

defines the information already available about a fragmented instance;

$H(\Pi \mid F, S)$

quantifies unresolved configurational uncertainty;

$H(Y \mid F, S)$

quantifies the uncertainty relevant to a specified global verification outcome;
the conditional feasible support is partitioned into validation-equivalence classes;

$\Gamma_{val} = H(\Pi \mid Y, F, S)$

quantifies configurational uncertainty that need not be resolved merely to determine that outcome; and additional execution-specific transcripts quantify the information that remains necessary when complete identification or task determination is required.
The novelty claim of GEDT is therefore made at the level of this **combined theoretical architecture and its proposed cross-domain interpretation**, rather than at the level of conditional entropy, mutual information, functional computation, statistical sufficiency, equivalence-class reduction, preprocessing, or amortized communication considered individually.

Table 1 - Positioning of GEDT relative to neighboring theoretical paradigms.

Prior paradigm	Established idea relevant to GEDT	GEDT-specific role
Side information / Slepian–Wolf	Available correlated information reduces residual coding uncertainty.	S is modeled as a persistent reusable structural core constraining feasible configurations across compatible executions.
Functional computation / characteristic-graph approaches	A target function may be computed without reconstructing the complete source.	GEDT embeds this distinction in a configurational model containing Π, S, validation-equivalence classes, residual configurational entropy, and separate reconstruction/validation transcripts.
Sufficient statistics / Blackwell sufficiency	A reduced representation may preserve all information required for inference or a decision task.	GEDT treats structural sufficiency as a property of a persistent system architecture and distinguishes state identification from global validation.
Information Bottleneck	A compressed representation may retain target-relevant information while discarding other state information.	GEDT does not optimize a representation; it quantifies how pre-existing structural constraints reshape the feasible configuration space and task-specific uncertainty.
Computational mechanics	Predictively equivalent histories can be grouped into minimal sufficient causal states.	GEDT groups feasible configurations by equality of a global validation outcome rather than histories by predictive distributions.
Algorithmic statistics	Information can be separated between a model/regularity description and residual specification.	GEDT uses Shannon conditional quantities to distinguish reusable structure, residual configuration uncertainty, and task-specific transcript requirements.
Knowledge compilation / information complexity	Preprocessing and information representation can reduce repeated query or amortized communication costs.	GEDT treats such savings as secondary consequences of structural epistemic sufficiency rather than as its central novelty.
Distributed snapshots / consensus	Global states or global decisions can be obtained through distributed information and coordination.	GEDT permits local components to remain epistemically limited while global validation is determined from local outputs plus structural constraints.
Order/lattice theory / Formal Concept Analysis	Partial orders, meets and shared attributes provide formal constructions of common structure.	GEDT uses the meet of structural descriptions over the admissible configuration family to construct s_0, then connects this core to residual configurational uncertainty and task-specific validation.
GEDT	—	Joint treatment of persistent structural information, residual configurational uncertainty, validation-equivalence, identification–validation separation, dynamic transcript cost, and secondary amortized consequences.

Mathematical model

Let:

- X be the global raw state of an instance;
- F = D(X) be the set of fragments generated by X;
- $\mathcal{P}$ be the finite set of global feasible configurations;
- Π be a random variable that takes values in $\mathcal{P}$ and represents the realized global configuration of the fragmented instance, including roles, ordering, relationships, and instance-specific associations;
- S be a structural variable taking values in the set $\mathcal{S}$ of admissible structural cores. When structural comparison is required, $\mathcal{S}$ is endowed with the partial order $\preceq$ defined below. Each realization s represents a reusable structural core containing feasibility constraints, invariants, structural relationships, role schemas, and decoding geometry. The same realization s can be reused across multiple compatible instances or executions. By

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definition, S does not encode Π as a complete and execution-specific explicit description;

• M_{Π} be an additional configurational transcript possibly required for the complete reconstruction of Π;

• $\mathcal{V}$ the deterministic global verification function;

• Y ∈ {ACCEPT, VETO} the outcome of the global verification, defined by $Y = \mathcal{V}(F, \Pi, S)$

• M_Y an additional transcript possibly required to precisely determine outcome Y;

• Π^* and Y^*, respectively, the reconstructed configuration and the reconstructed outcome of the global verification.

It is also assumed that fragmentation is informationally reversible once the correct configuration is known:
$H(X \mid F, \Pi) = 0$.

Equivalently, there exists a reconstruction function $\mathcal{R}$ such that:
$X = \mathcal{R}(F, \Pi)$
almost surely.

Configurational reconstruction and global verification

The complete configurational reconstruction is defined by:
$\Pi^* = g_{\Pi}(F, S, M_{\Pi})$ *(1)*,
with the exactness requirement:
$P(\Pi^* = \Pi) = 1$ *(2)*.
Equivalently:
$P(\Pi^* \neq \Pi) = 0$ *(3)*.
Since fragmentation is reversible with respect to the correct configuration, the exact reconstruction of Π also allows the lossless reconstruction of the global raw state:
$X^* = \mathcal{R}(F, \Pi^*)$
almost surely.

The outcome of the global verification can instead be reconstructed directly by:
$Y^* = g_Y(F, S, M_Y)$ *(4)*,
with:
$P(Y^* = Y) = 1$.

Since Y is a deterministic function of F, Π, and S, the following holds:
$H(Y \mid F, S) \leq H(\Pi \mid F, S)$ *(5)*.

Consequently, exact configurational reconstruction is sufficient to determine the outcome of the global verification, but it is not necessarily required in every architecture. A structural core may be sufficient to produce the exact outcome of the global verification even when it does not uniquely identify the entire implemented configuration.

Proposition 1 — Validation–Identification Separation

Note
For Proposition 1, F and S are treated as discrete variables, consistently with the finite fragmented-system setting considered here.
The following characterization follows from standard properties of conditional entropy and deterministic mappings.
Its role in GEDT is to formalize the distinction between complete configurational identification and task-specific global validation.

Let F denote the observed fragments, S the reusable structural core, Π the realized global configuration, and

$Y = \mathcal{V}(F, \Pi, S)$

a deterministic global verification outcome.

For every pair (f, s) having positive probability, define the conditional feasible support

$P_{f,s} := \mathrm{supp}(\Pi \mid F=f, S=s)$.

Define on $P_{f,s}$ the validation-equivalence relation

$\pi \sim_{f,s} \pi' \iff \mathcal{V}(f, \pi, s) = \mathcal{V}(f, \pi', s)$.

Then:

(i) Configurational identification.
The global configuration is exactly identifiable from F and S,

$H(\Pi \mid F, S) = 0$,

if and only if $P_{f,s}$ contains a single configuration for every pair (f, s) such that $P(F=f, S=s) > 0$.

(ii) Global validation.
The verification outcome is exactly determined by F and S,

$H(Y \mid F, S) = 0$,

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if and only if, for every pair (f, s) such that P(F=f, S=s) > 0, all configurations in $P_{f,s}$ belong to the same validation-equivalence class.
Equivalently,

$\mathcal{V}(f, \pi, s)$

is constant over $P_{f,s}$.

(iii) Strict validation–identification separation.
Consequently, exact global validation without complete configurational identification occurs whenever

$H(Y \mid F, S) = 0$ and $H(\Pi \mid F, S) > 0$.

In this regime, multiple global configurations remain epistemically compatible with the available fragments and structural core, but all of them induce the same global verification outcome.

Proof

Since Π is discrete, $H(\Pi \mid F, S) = 0$ if and only if Π is almost surely a deterministic function of (F, S). This is equivalent to the conditional support $P_{f,s}$ containing a single configuration for every pair (f, s) such that P(F=f, S=s) > 0.

Because $Y = \mathcal{V}(F, \Pi, S)$ is deterministic once F, Π, and S are specified, $H(Y \mid F, S) = 0$ if and only if Y is almost surely a deterministic function of (F, S).
For a fixed (f, s), this occurs exactly when every configuration π having positive conditional probability under $\Pi \mid F=f, S=s$ produces the same value of $\mathcal{V}(f, \pi, s)$.
Hence all configurations in $P_{f,s}$ must belong to the same validation-equivalence class.

Therefore, singleton conditional support is sufficient for exact validation but is not necessary. A non-singleton conditional support may remain while the verification map is constant over that support, yielding

$H(Y \mid F, S) = 0$ with $H(\Pi \mid F, S) > 0$.

□

Definition — Validation Gap

Define the validation gap as

$\Gamma_{val} := H(\Pi \mid F, S) - H(Y \mid F, S)$.

Since Y is a deterministic function of (F, Π, S).

$H(\Pi, Y \mid F, S) = H(\Pi \mid F, S)$

and, by the conditional chain rule,

$H(\Pi, Y \mid F, S) = H(Y \mid F, S) + H(\Pi \mid Y, F, S).$

Therefore,

$\Gamma_{val} = H(\Pi \mid Y, F, S) \geq 0$

and Γ_{val} quantifies the residual configurational uncertainty that need not be resolved merely to determine the global verification outcome.

In the strict separation regime,

$H(Y \mid F, S) = 0, H(\Pi \mid F, S) > 0,$

one obtains

$\Gamma_{val} = H(\Pi \mid F, S) > 0.$

Example — Equal configurational uncertainty, different validation sufficiency

Consider two fragmented systems for which, conditional on the available fragments and structural core, two equiprobable global configurations remain possible. In both systems,

$H(\Pi \mid F, S) = 1$ bit.

In the first system, both admissible configurations produce the same verification outcome:

$\mathcal{V}(f, \pi_1, s) = \mathcal{V}(f, \pi_2, s) = \text{ACCEPT}.$

Hence

$H(Y \mid F, S) = 0$

despite the remaining configurational uncertainty.

In the second system,

$\mathcal{V}(f, \pi_1, s) = \text{ACCEPT}, \mathcal{V}(f, \pi_2, s) = \text{VETO},$

and therefore

$H(Y \mid F, S) = 1$ bit.

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The two systems have the same residual configurational uncertainty but radically different epistemic sufficiency with respect to global validation.
GEDT therefore distinguishes not only **how much** configurational uncertainty remains, but also **whether that uncertainty is relevant to the global task being evaluated**.

Structural Sufficiency and Implicit Metadata

If a structural core S makes the global configuration Π determinable from fragments F, then the order, roles, relationships, and constraints already inferable from (F, S) need not be re-specified by an additional execution-specific transcript.

Thus, assuming that S is already available to the global reconstructor, if $H(\Pi \mid F, S) = 0$, the additional configurational transcript may be chosen empty, $M_\Pi = \emptyset$, and therefore $\ell(M_\Pi) = 0$.

This result does not imply that the construction, distribution, or maintenance of S is costless.

Quantitative consequence for configurational reconstruction

The preceding characterization distinguishes the information required to identify the complete global configuration from the information required to determine the global verification outcome.
When complete configurational reconstruction is nevertheless required, the residual uncertainty $H(\Pi \mid F, S)$ imposes a corresponding informational burden on any additional execution-specific transcript.

The following theorem quantifies this burden using standard information-theoretic bounds. It should therefore be interpreted as a quantitative consequence of residual configurational uncertainty, rather than as the primary conceptual distinction introduced by GEDT.

Theorem 2 — Residual informational cost of configurational reconstruction

The following theorem extends the previous lossless model to a general regime with bounded error; the exact case is recovered by setting $\varepsilon_\Pi = 0$.

Definition 1 — Feasible Configuration Space

Let Π be a discrete random variable that takes values in the finite set of admissible configurations $\mathcal{P}$, with $|\mathcal{P}| \geq 2$.

Definition 2 — Reusable Structural Core

Let F be the set of observed fragments,
S be a structural variable that takes values in the set $\mathcal{S}$ of admissible structural cores.
Let
M_Π be an additional configurational transcript represented by a binary encoding that is uniquely decodable given F and S.

Definition 3 — Residual Configurational Uncertainty

Define the residual configurational uncertainty:
$H_{res} := H(\Pi \mid F, S)$ *(6)*.
Let
$\Pi^* = g_\Pi(F, S, M_\Pi)$ *(7)*
be a reconstruction of the global configuration such that:
$P(\Pi^* \neq \Pi) \leq \varepsilon_\Pi$ *(8)*
with $0 \leq \varepsilon_\Pi \leq 1$,
and let
$p_\Pi := P(\Pi^* \neq \Pi)$ and therefore
$p_\Pi \leq \varepsilon_\Pi$.

Definition 4 — Fano Correction Term

Let us define $\delta_\mathcal{P}$ the Fano informational term, which upper bounds the residual uncertainty compatible with an error probability ε_Π:
$\delta_\mathcal{P}(\varepsilon_\Pi) := h_2(\bar{\varepsilon}_\Pi) + \bar{\varepsilon}_\Pi \log_2 (|\mathcal{P}| - 1)$ *(9)*,
with
$\bar{\varepsilon}_\Pi := \min \{\varepsilon_\Pi, 1 - (1/ |\mathcal{P}|)\}$ *(10)*.

Let
$h_2(p) = -p \log_2 p - (1 - p) \log_2 (1 - p)$ *(11)*
the binary entropy relating to the event
$E = 1_{\{\Pi \wedge * \neq \Pi\}}$,
i.e., error/no error.

Since the function
$f(p) = h_2(p) + p \log_2 (|\mathcal{P}| - 1)$ *(12)*
increases up to
$p^* = 1 - (1/ |\mathcal{P}|)$,
after this point, it decreases.
Therefore, the maximum of f(p) for all actual error probabilities consistent with $p \leq \varepsilon_\Pi$ is obtained in $\bar{\varepsilon}_\Pi$.

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Then the average length of the additional transcript satisfies:
$\mathbb{E}[\ell(M_\Pi)] \geq \max\{0, H_{res} - \delta_{\mathcal{P}}(\varepsilon_\Pi)\}$ *(13)*.

Definition 5 — Epistemic Dispersion

Define
$D_{epi}(S ; \Pi \mid F) := I(\Pi ; S \mid F)$ *(14)*
and therefore:
$D_{epi}(S ; \Pi \mid F) = H(\Pi \mid F) - H(\Pi \mid F, S)$ *(15)*.
The central decomposition follows:
$H(\Pi \mid F) = D_{epi}(S ; \Pi \mid F) + H_{res}$ *(16)*.

Interpretation:
• $H(\Pi \mid F)$, initial configurational entropy;
• D_{epi}, part absorbed or reallocated in the structure;
• H_{res}, part that remains to be resolved dynamically.

Exact reconstruction
In case:
$\varepsilon_\Pi = 0$,
you have:
$\delta_{\mathcal{P}}(0) = 0$,
and therefore:
$\mathbb{E}[\ell(M\Pi)] \geq H_{res}$.

Non-identifying regime in exact reconstruction
When:
$H_{res} > 0$,
the transcript cannot be identically empty and its average length must be strictly positive:
$\mathbb{E}[\ell(M_\Pi)] \geq H_{res} > 0$ *(17)*.
Identifying regime
When:
$H_{res} = 0$,
the configuration Π is almost surely determined by F and S.
There therefore exists a function g_0 such that:
$\Pi = g_0(F, S)$ almost surely.

In this case, the additional configurational transcript can be empty:
$M_\Pi = \emptyset$, and the reconstruction:
$\Pi^* = g_0(F, S)$
satisfies:
$P(\Pi^* = \Pi) = 1, \ell(M_\Pi) = 0$ *(18)*.

Proof

Since M_Π is represented by a conditionally uniquely decodable binary encoding and the reconstructor has F and S, the source coding theorem implies:
$\mathbb{E}[\ell(M_\Pi)] \geq H(M_\Pi \mid F, S)$.
By the non-negativity of the conditional entropy:
$H(M_\Pi \mid F, S) \geq I(\Pi ; M_\Pi \mid F, S)$,
since
$I(\Pi ; M_\Pi \mid F, S) = H(M_\Pi \mid F, S) - H(M_\Pi \mid \Pi, F, S)$.
By definition of conditional mutual information:
$I(\Pi ; M_\Pi \mid F, S) = H(\Pi \mid F, S) - H(\Pi \mid F, S, M_\Pi)$ *(19)*.

Since $\Pi^* = g_\Pi(F, S, M_\Pi)$, the reconstruction Π^* is a function of the information available to the reconstructor.

Therefore:
$H(\Pi \mid F, S, M_\Pi) \leq H(\Pi \mid \Pi^*)$.
Applying the Fano inequality:
$H(\Pi \mid \Pi^*) \leq h_2(p_\Pi) + p_\Pi \log_2 (|\mathcal{P}| - 1) \leq \delta_{\mathcal{P}}(\varepsilon_\Pi)$ *(20)*.

Consequently:
$H(\Pi \mid F, S, M_\Pi) \leq \delta_{\mathcal{P}}(\varepsilon_\Pi)$.
Substituting:
$I(\Pi ; M_\Pi \mid F, S) \geq H_{res} - \delta_{\mathcal{P}}(\varepsilon_\Pi)$,
and therefore:
$\mathbb{E}[\ell(M_\Pi)] \geq H_{res} - \delta_{\mathcal{P}}(\varepsilon_\Pi)$ *(21)*.

Since an average length cannot be negative, the limit can be expressed as:
$\mathbb{E}[\ell(M_\Pi)] \geq \max\{0, H_{res} - \delta_{\mathcal{P}}(\varepsilon_\Pi)\}$.
In the exact case, $\varepsilon_\Pi = 0$, the Fano term is zero and we obtain:
$\mathbb{E}[\ell(M_\Pi)] \geq H_{res}$ *(22)*.

Finally, when $H_{res} = 0$, the discrete nature of Π implies that the configuration is almost surely determined by F and S. Therefore, an exact reconstruction exists without additional configurational transcript.

The theorem concerns complete configurational reconstruction. It does not imply that resolving H_{res} is necessary for every global task. As established by the validation–identification characterization, a verification outcome may already be exactly determined even when $H_{res} > 0$, provided the remaining feasible configurations are equivalent with respect to the verification function.

Interpretation

Structural constraints can function as implicit, reusable metadata.
The reduction in residual configurational uncertainty does not represent disappearing

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information; it reflects information supplied by S rather than information that must be re-specified by an execution-specific transcript.

Canonical definition of the reusable structural core

In the general GEDT framework, S represents the persistent structural information that constrains the set of global configurations compatible with a fragmented instance. This structural core can be given a constructive definition by interpreting it as the greatest common structural component shared by all configurations admitted by the operational constraints.

Let

$\mathcal{X}_n = \{0, 1\}^n$

be the space of all possible configurations of a binary system, and let O denote the operational constraints. The corresponding admissible configuration set is

$\mathcal{P}_0 := \{\pi \in \mathcal{X}_n : O(\pi) = 1\};$

In this constructive binary specialization, $\mathcal{P}_0$ instantiates the feasible configuration set $\mathcal{P}$ of the general GEDT model (the order-theoretic construction uses the standard notions of partial order and greatest lower bound (Davey & Priestley, 2002)).

Each realization

$\pi = (x_1, x_2, \ldots, x_n) \in \mathcal{P}_0$

specifies the local state of the n system components.

Let $\mathcal{S}$, already introduced as the set of admissible structural cores, be endowed with a partial order $\preceq$, where $s_1 \preceq s_2$ means that the structural information represented by s_1 is contained in, or implied by, s_2.

Let

$\sigma : \mathcal{P}_0 \to \mathcal{S}$

associate each admissible configuration with its structural description.
The mapping σ and the partial order $\preceq$ must be specified independently of the particular realized configuration; consequently, the canonicality of s_0 is relative to the declared structural representation $(\mathcal{S}, \preceq, \sigma)$.

Let

$s_0 \in \mathcal{S}$

be the canonical core, a distinguished element of the structural-core space $\mathcal{S}$, determined by the admissible configuration family associated with O.

Whenever the corresponding greatest lower bound exists, GEDT defines the canonical reusable structural core as

$$s_O := \bigwedge_{\pi \in \mathcal{P}_o} \sigma(\pi)$$

or equivalently,

$s_0 = \max_{\preceq} \{s \in \mathcal{S} : s \preceq \sigma(\pi)\ \forall \pi \in \mathcal{P}_0\}$.

Thus s_0 is the greatest structural object contained in every structural description $\sigma(\pi)$ associated with an admissible configuration $\pi \in \mathcal{P}_0$: the structural analogue of a greatest common divisor. It contains only relations, invariants, roles, or constraints that survive across the entire admissible configurational family and does not encode the execution-specific identity of any single configuration.

The use of shared attributes and lattice-theoretic structure has established precedents, notably in Formal Concept Analysis (Ganter & Wille, 1999); GEDT does not claim the meet construction itself as novel, but uses it here to construct the persistent structural core entering the configurational-information decomposition.

The definition makes an important distinction between structural invariance and configurational identification.

Several distinct configurations may possess the same canonical structural core, so that $H(\Pi \mid F, S=s_0) > 0$ can remain positive even though s_0 is completely specified. Nevertheless, if all configurations remaining compatible with F and s_0 induce the same global validation outcome Y, then $H(Y \mid F, S=s_0) = 0$ may coexist with $H(\Pi \mid F, S=s_0) > 0$.

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The canonical structural core therefore identifies the maximum reusable structural component common to the admissible configuration family, while the residual conditional entropy measures the information that remains instance-specific. Across a family of architectures S may vary, whereas within repeated executions of a given architecture the same realization S = s may persist and be reused. This preserves the GEDT distinction between persistent structural information and execution-specific configurational information.

The epistemic weight of global understanding

Each node that must autonomously determine global integrity must receive, possess, or reconstruct enough information to resolve its uncertainty about the outcome of the global verification to handle partial data.

Let:

- $Y \in \{ACCEPT, VETO\}$, the outcome of the global verification;
- V_i, the local knowledge available to node i.

Let $Y_i^* = g_i(V_i, M_i)$ be an estimator of the global verification outcome,
with:
$P(Y_i^* \neq Y) \leq \varepsilon$.

If the node must calculate Y using a message M_i, with error at most ε, then:
$I(Y ; M_i \mid V_i) \geq H(Y \mid V_i) - h_2(\varepsilon)$,
with $0 \leq \varepsilon \leq 1/2$.

Consequently, assuming a binary encoding of M_i that is uniquely decodable given V_i:
$\mathbb{E}[\ell(M_i)] \geq H(M_i \mid V_i) \geq I(Y ; M_i \mid V_i)$ *(23)*.

Therefore:
$\mathbb{E}[\ell(M_i)] \geq H(Y \mid V_i) - h_2(\varepsilon)$ *(24)*.

If, however, the node must only perform a local measurement:
$L_i = f_i(V_i)$,
the global outcome is produced by the deterministic global verification function:
$Y = V_G(L_1, \ldots, L_n ; S)$.
Then the node does not need to receive the information needed to understand Y, Π, or the entire X.

Information complexity theory generally confirms that distributed computation of a function can require the revelation of a positive amount of information, even independently of the simple number of rounds (Kushilevitz & Nisan, 1997; Braverman & Rao, 2014).

Corollary — Residual Informational Cost of the Global Verification Outcome

Assume that M_Y is represented by a conditionally uniquely decodable binary encoding given F and S.
If:
$Y^* = g_Y(F, S, M_Y)$
and
$P(Y^* \neq Y) \leq \varepsilon_Y$,
then, since Y is binary:
$E[\ell(M_Y)] \geq \max\{0, H(Y \mid F, S) - h_2(\varepsilon_Y)\}$ *(25)*,
with
$0 \leq \varepsilon_Y \leq 1/2$.
In the exact regime:
$\varepsilon_Y = 0$,
we obtain:
$E[\ell(M_Y)] \geq H(Y \mid F, S)$.
If
$H(Y \mid F, S) = 0$,
the outcome of the verification is already determined by F and S, even when:
$H(\Pi \mid F, S) > 0$.

Economic-structural corollary as a secondary implication

Note
All costs are expressed relative to a common cost function or mapped to a common metric, such as economic cost, normalized energy consumption, computational time, or an explicitly stated weighted combination.

The following cost relation is not proposed as the primary theoretical contribution of GEDT. It is a secondary system-level implication that becomes relevant only when structural epistemic sufficiency actually reduces the amount of execution-specific information, communication, coordination, or semantic replication required by an implementation. The informational results of GEDT do not by themselves guarantee an economic, energetic, or computational advantage.

Let C_S be the cost of building, deploying, and maintaining structure S.
Let C_E be the initial cost of the semantically explicit architecture.
Let $C_{local,\,t}^{S}$, and $C_{local,\,t}^{E}$ be the respective local costs for execution at cycle t, and let $C_{add,\,t}$ be the additional cost of metadata, semantic replication, coordination required by the explicit architecture for execution cycle t, where t serves as a generic execution index.

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After Z executions:

$C_{struct} = C_S + \sum_{t=1}^{Z} \left(C^S_{local,t} \right)$, *(26)*

While, for the explicit architecture:

$C_{explicit} = C_E + \sum_{t=1}^{Z} \left(C^E_{local,t} + C_{add,t} \right)$ *(27)*.

Let

$\Delta C_t := \left(C^E_{local,t} + C_{add,t} \right) - C^S_{local,t}$

where ΔC_t represents the cost difference between the explicit architecture and the structural architecture during execution at cycle t.

The structure is overall less expensive when:

$C_s - C_E < \sum_{t=1}^{Z} \Delta C_t$. *(28)*

In the simplified case where $C_E = 0$,

the condition boils down to:

$C_S < \sum_{t=1}^{Z} \left(\Delta C_t \right)$ *(29)*

If the average savings per execution is $\Delta C^* > 0$, the theoretical payback point is:

$Z > C_S / \Delta C^*$ *(30)*.

Accordingly, the economic condition should be read as a contingent consequence of structural reuse under an explicitly specified cost model, not as evidence for the novelty or validity of the underlying epistemic framework.

Interpretative limits

The results establish informational bounds conditional on the probabilistic distribution considered and the declared decodability assumptions. They do not automatically demonstrate a universal reduction in energy, latency, or cost in every single implementation. Such advantages require that the cost of building and maintaining the framework be lower than the cost associated with the explicit information and semantic replication thereby avoided.

Conclusion

GEDT distinguishes complete configurational identification from epistemic sufficiency for a global task. A persistent structural core can restrict the set of feasible global configurations without necessarily identifying a unique configuration, and exact global verification can remain possible when the unresolved configurations are equivalent with respect to the verification function.

Residual configurational uncertainty determines the additional informational burden required when full reconstruction is demanded, whereas only task-relevant uncertainty must be resolved when the objective is global validation.

Structural reuse may consequently reduce communication, coordination, or amortized system costs in suitable implementations, but such advantages are secondary and depend on the adopted architecture and cost model.

Data Availability Statement

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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The Author acknowledges the use of AI tools for language translation and refinement, editorial revision, consistency checking, and assistance in mathematical formalization and verification of derivations. The conceptual architecture, research questions, theoretical framework, intended interpretation of the formal model, and scientific claims were conceived, directed, evaluated, and approved by the Author. The Author has reviewed the entire manuscript and assumes full scientific responsibility for its content.

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